

ADVANCED MICROECONOMETRICS
Problem Set - Day 4

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1. You already estimated the Roy model using a parametric approach. Now the idea is to apply nonparametric techniques to the estimation of the Marginal Treatment Effect. To this end, use the model from Problem Set 3, but assume $\beta_0 = \beta_1 = 0$.

- (a) Show that

$$E(Y|P(Z) = p) = \alpha + K(p), \quad (1)$$

where $P(Z)$ is the propensity score, or probability of selection, p is a particular evaluation value of the propensity score, and

$$K(p) = \varphi p + E(U_0|P(Z) = p) + E(U_1 - U_0|D = 1, P(Z) = p)p. \quad (2)$$

- (b) Use expressions (1) and (2) to explain the steps involved in the semiparametric estimation of the Marginal Treatment Effect as presented in Heckman, Urzua and Vytlacil (2006).
- (c) Explain in detail why you can approximate $K(p)$ with a polynomial in p . Explain how you can estimate the Marginal Treatment Effect using this approximation.
- (d) Estimate the Marginal Treatment Effect using *LIV* and the polynomial approximation of question (c). Graph the estimated Marginal Treatment Effect for each approach (including its confidence interval) against its actual value. Provide a brief explanation of the results you present.
- (e) Compute *TT*, *TUT*, and *LATE* (select two values of p) and compare them with their actual values. Present a figure comparing the actual and estimated weights of *TT* and *TUT*.
- (f) Consider what *IV* estimates when Z_1 and the propensity score are used as instruments. Explain why the propensity score can be used as an instrument. Show the associated weights. Specifically, derive the IV weights for both Z_1 and P . Plot

the density of P and the joint density of Z_1, P . Plot the IV weights as a function of u_D :

$$\frac{E(Z_1 - \bar{Z}_1 \mid P(Z) > u_D) \Pr(P(Z) > u_D)}{Cov(Z_1, D)}$$

for both sets of weights.

- (g) Now implement the nonparametric version of the MTE. For this,
 - i. Examine whether you can verify the common support assumption.
 - ii. Select the values of the propensity score (grid).
 - iii. Select your kernel and implement an algorithm for bandwidth selection.
 - iv. Implement the local linear regression approach discussed in class.
 - v. Display the true MTE against your estimated values.
 - vi. Estimate confidence intervals by bootstrapping.
- 2. Propose your own three-sector model and implement the IV method discussed in class.